

FRC 2023 CHARGED UP — Top Robot Design Deep Dive

A study reference for effective CHARGED UP design & implementation, drawn from the Spectrum 3847 CAD Collection, team CAD/code releases (Onshape / GrabCAD), and each team's Chief Delphi reveal / CAD-release thread.

The CHARGED UP design landscape

The game

CHARGED UP is a cone-and-cube game: robots collect CONES and CUBES from the floor or the SINGLE/DOUBLE SUBSTATIONS and score them on the GRID's three rows (hybrid, mid, high) for links and a co-op bonus, then in the endgame DOCK and ENGAGE on the CHARGE STATION. Most elite designs paired a fast cone/cube cycle (an arm or elevator + a cone-and-cube-capable end effector) with reliable auto-balance; many top teams ran a telescoping or double-jointed arm and treated triple-balance/links as the high-value targets.

How to use this document

These are the most thoroughly-documented CHARGED UP robots from the CAD-release set, each with real subsystem detail mined from the team's Chief Delphi reveal/release thread. For every team we capture archetype, the four subsystems (shooter / intake / indexer / drivetrain), vision, public CAD availability, and 'lessons for students'. ★ marks the curated study picks with the richest reproducible documentation. EPA / rank are pending a Statbotics API outage and will be backfilled when it recovers.

Profiled robots at a glance

Team	Robot	Local EPA	Rec	Archetype	Public CAD/Design
★ 1678 Citrus Circuits	Tangerine Tumbler	—	—	Strong elevator cone/cube robot; full CAD + code release	CD thread
★ 971 Spartan Robotics	Doffle	—	—	Tank-drive cone/cube robot — kept tank in a swerve-dominated y	Onshape CAD
★ 6328 Mechanical Advantage	(2023)	—	—	Open-Alliance, software-led; reliable-and-simple philosophy; A	CD thread
★ 2910 Jack in the Bot	Phantom	—	—	Strong PNW cone/cube robot; CAD + tech binder	CD thread
★ 1114 Simbotics	Simbot Scizor	—	—	Design-process leaders; CAD + code + reusable base	CD thread
★ 695 Bison Robotics	Goldmodule	—	—	Swerve cone/cube robot; public code + a swerve-CAD teaching re	CD thread
★ 3005 RoboChargers	AMP	—	—	Cone/cube robot with a widely-praised end effector	Onshape CAD
★ 2767 Stryke Force	Snake Eyes	—	—	Cone/cube robot with an operator-friendly vision-align indicat	Onshape CAD
★ 1706 Ratchet Rockers	Whiplash	—	—	Cone/cube robot with a grid-registering frame	Onshape CAD
★ 118 Robonauts	Echo	—	—	Cone/cube robot; also the FRC Everybot authors	GrabCAD

★ = recommended study pick (downloadable CAD or a full technical binder + a clear design lesson). EPA shown is Local EPA (approx), computed from The Blue Alliance match data.

Team-by-team profiles

★ Study pick

1678 Citrus Circuits Tangerine Tumbler
USA-CA

EPA pending

Strong elevator cone/cube robot; full CAD + code release

SCORING (CONE / CUBE) Cone/cube end effector for grid scoring (detail in released CAD).	INTAKE Cone/cube intake feeding the elevator (see CAD).
ARM / ELEVATOR Elevator superstructure (cable/spool-tensioned; discussed in the release Q&A).	DRIVETRAIN Swerve.
ENDGAME Charge Station: drive on and auto-balance (engage) — no dedicated climber; a drivetrain + controls task.	AUTONOMOUS Multi-game-piece autos ending in a Charge Station engage (see released code).
VISION Unconfirmed (check code)	PUBLIC CAD / DESIGN CD CAD & code release

LESSONS FOR STUDENTS

- A top team's full CAD + code, with a public robot-history archive going back to 2009 — study how a program iterates year over year.
- Cable/spool elevator tensioning is a classic gotcha worth understanding from their Q&A.

CD thread: <https://www.chiefdelphi.com/t/437632> · CD CAD & code release · CD reveal

★ Study pick

971 Spartan Robotics Doffle
USA-CA

EPA pending

Tank-drive cone/cube robot — kept tank in a swerve-dominated year by choice

SCORING (CONE / CUBE) Cone/cube scoring end effector (detail in released CAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN 6-wheel TANK / west-coast drive — a deliberate choice to keep the robot to a reasonable number of degrees of freedom while everyone else went swerve.
ENDGAME Charge Station dock & engage; tank drive gives a stable, repeatable straight drive-on.	AUTONOMOUS Preload + pickup, then Charge Station balance (see CAD/code).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- An experienced team deliberately staying with tank drive is a useful lesson in choosing simplicity / fewer DOFs over the trendy option.
- 971's mechanical execution and packaging are worth studying regardless of drivetrain.

Onshape CAD: <https://frc971.onshape.com/documents/db1b739c445de530fb406dc6/v/bc9815d549b9ffd9f7251ac8/e/e81d7e62d16fb20bf281c8e4> · CD reveal — Doffle

★ Study pick

6328 Mechanical Advantage (2023)

USA-MA

EPA pending

Open-Alliance, software-led; reliable-and-simple philosophy; AdvantageKit authors

SCORING (CONE / CUBE) Cone/cube end effector (mechanism detail in released CAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve.
ENDGAME Charge Station auto-balance, controller tuned against AdvantageKit logs.	AUTONOMOUS Reliable preload-and-balance autos; log-based tuning.
VISION Advanced logging + vision via their own AdvantageKit framework (see code).	PUBLIC CAD / DESIGN CD 2023 Open-Alliance build thread

LESSONS FOR STUDENTS

- 6328 author AdvantageKit — the reference FRC logging/replay framework; their code is the model for log-based development.
- Explicit 'reliable, relatively simple, lean into software strengths' design philosophy + full-team prototyping is worth emulating.

CD thread: <https://www.chiefdelphi.com/t/420691> · CD 2023 build thread · Onshape CAD

★ Study pick

2910 Jack in the Bot Phantom

USA-WA

EPA pending

Strong PNW cone/cube robot; CAD + tech binder

SCORING (CONE / CUBE) Cone/cube scoring end effector (full detail in the tech binder).	INTAKE Cone/cube intake (tech binder).
ARM / ELEVATOR Arm/elevator superstructure (tech binder).	DRIVETRAIN Swerve.
ENDGAME Charge Station dock & engage.	AUTONOMOUS Multi-piece scoring then balance (tech binder).
VISION Unconfirmed (check binder)	PUBLIC CAD / DESIGN CD CAD & tech binder release

LESSONS FOR STUDENTS

- Ships both Onshape and STEP/Parasolid plus a tech binder — a complete, clone-friendly documentation set.
- A well-documented benchmark PNW robot to compare against the elevator/arm field.

CD thread: <https://www.chiefdelphi.com/t/436653> · CD CAD & tech binder

★ Study pick

1114 Simbotics Simbot Scizor

CAN-ON

EPA pending

Design-process leaders; CAD + code + reusable base

SCORING (CONE / CUBE) Cone/cube scoring end effector (detail in released CAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve.
ENDGAME Charge Station auto-balance.	AUTONOMOUS Preload + balance routines; Simbot-Base utilities.
VISION Unconfirmed (check code)	PUBLIC CAD / DESIGN CD CAD & code release

LESSONS FOR STUDENTS

- Released the robot CAD plus Simbot-Base — a maintained starter code base with up-to-date utilities to bootstrap a season fast.
- 1114 are long-standing design-process mentors; study their packaging and conventions.

CD thread: <https://www.chiefdelphi.com/t/447607> · CD CAD & code release · Code — Simbotics/2023-Simbot-Scizor

Swerve cone/cube robot; public code + a swerve-CAD teaching resource

SCORING (CONE / CUBE) Cone/cube scoring end effector (see CAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve; code developed in public, conventions carried from Rapid React.
ENDGAME Charge Station dock & engage on swerve.	AUTONOMOUS Preload + pickup + balance (public code).
VISION Offseason vision/odometry code rewrite (see GitHub).	PUBLIC CAD / DESIGN CD CAD & code release

LESSONS FOR STUDENTS

- They published a parametric, template-part-based MK4i swerve SOLIDWORKS tutorial series — an outstanding learn-to-CAD-swerve resource.
- Developing code in public + documenting offseason swerve/vision rewrites is a model for continuous learning.

CD thread: <https://www.chiefdelphi.com/t/446357> · [CD CAD & code release](#) · [Code](#) — [FRCTeam695](#)

Cone/cube robot with a widely-praised end effector

SCORING (CONE / CUBE) A cone/cube end effector that drew a lot of community praise on reveal ('Holy end effector!') — study its grip/geometry in the released CAD.	INTAKE Cone/cube intake integrated with the end effector (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve.
ENDGAME Charge Station dock & engage.	AUTONOMOUS Cone/cube preload autos; balance (specifics unconfirmed).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- A widely-noted end effector — a great single-mechanism study for handling both cones and cubes.
- Reverse-engineer the released Onshape to see how one gripper manages two very different game pieces.

Onshape CAD: <https://cad.onshape.com/documents/230339548ba1828471e00cef/w/7608c62ea34bc84ee4acccd6/e/6a002e572ec5e1812bae3918> · [CD reveal](#) — AMP

Cone/cube robot with an operator-friendly vision-align indicator

SCORING (CONE / CUBE) Cone/cube scoring end effector (see CAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve.
ENDGAME Charge Station auto-balance; the vision-align indicator helps pre-balance scoring placement.	AUTONOMOUS Preload + balance (specifics unconfirmed).
VISION Camera auto-align to the grid, with a light-up indicator that turns on when the camera alignment engages — a clean driver-feedback idea.	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- A visible status light tied to vision auto-align is a simple, high-value human-factors touch worth copying.
- Study how the alignment indicator integrates with the scoring cycle.

Onshape CAD: <https://cad.onshape.com/documents/04a54b2035c31e128e1dfced/w/505caf917c2302fdb0b5bdd9/e/4c7eedc3e2c9f9a4836827bf> · CD reveal — Snake Eyes

Cone/cube robot with a grid-registering frame

SCORING (CONE / CUBE) Cone/cube scoring end effector (see CAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve.
ENDGAME Charge Station dock & engage; the octagonal frame aids grid registration before crossing to the platform.	AUTONOMOUS Preload + balance (specifics unconfirmed).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- A slightly OCTAGONAL frame perimeter lets them jam the bumper into the grid to register/line up for scoring — a clever passive-alignment trick.
- Study how frame geometry can do alignment work that would otherwise need sensors.

Onshape CAD: <https://cad.onshape.com/documents/80bfb649a2c78e4c6edec3d3/w/6b74b6511ba0a48d4efc8200/e/9dac5cbb17ec4689a721dff4> · CD reveal — Whiplash

Cone/cube robot; also the FRC Everybot authors

SCORING (CONE / CUBE) Cone/cube scoring end effector (detail in the released GrabCAD).	INTAKE Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	DRIVETRAIN Swerve.
ENDGAME Charge Station dock & engage; Everybot-style accessible approach.	AUTONOMOUS Preload + balance; routines mirror the accessible Everybot playbook.
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN GrabCAD — Echo

LESSONS FOR STUDENTS

- 118 lead the Everybot program — the canonical low-cost, accessible CHARGED UP design many rookies build.
- Compare their full comp robot 'Echo' against the simplified Everybot to see design trade-offs.

GrabCAD: <https://grabcad.com/library/robonauts-118-2023-robot-echo-1> · [GrabCAD — Echo](#) · [CD reveal video](#)

Additional well-documented robots for study

Each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. ★ marks the recommended study picks.

Sources: the Spectrum 3847 CAD Collection and team CAD releases (Onshape, GrabCAD), Chief Delphi reveal / CAD-release threads, and match data from The Blue Alliance. EPA shown is Local EPA (approx), independently computed — not official Statbotics. "Unconfirmed" means a detail was not publicly published, not that it is absent.